

We used the Q-Potts cellular model in 2D with the total energy of a configuration given by two terms:

- the coupling energy between neighboring cells
- the area elasticity energy.

Starting with normal young and old RPE, we considered three repair mechanisms in different scenarios. The three mechanisms are:

1- Purse-String Closure:

a) After Cell Apoptosis: A cell is chosen to undergo apoptosis by setting its target volume to zero. Simultaneously, all the cells contacting this dead cell (nearest neighbors or NN) can enlarge according to $A_{final} = A_{initial} + A_{dead}$ cell number of NN cells $A_{final} = A_{initial} + \text{number of NN cells} \times A_{dead}$ cell. Thus, the NN cells stretch to cover the area of the dead cell, mimicking the purse-string closure.

b) After Cluster of Cell Apoptosis: A cluster of cells die, and in this case, the enlargement extends beyond the first nearest neighbors.

2- Cell Fusion:

a) Random Fusion: One cell is chosen at random, and one of its nearest neighbors adopts its ID, causing the two cells to merge into one with the same ID. Another fusion event can occur in two ways: either one of the neighbors joins this fused cell, or a new cell at random is chosen to fuse with its neighbor. Repeated fusion can happen up to 9 times, as the maximum number of nuclei observed in RPE cells is 10.

b) Wound-Induced Fusion: Following the concept from the manuscript "Wound-Induced Syncytia Outpace Mononucleate Neighbors during Drosophila Wound," a wound gradually closes, and 46% of the cells within five layers around the wound fuse, with 11 shrinking cells dying.

c) Chain Fusion: Following b), a chain of fusion is generated throughout the tissue.

3 -Cell Expansion:

A cell expands up to 5 times its area, while its first, second, and third nearest neighboring cells shrink by 20%, 15%, and 5% of their original areas, respectively. Additionally, any neighboring cell that reduces by more than 20% of its original area dies. The maximum expansion a cell can achieve is five times its original size.

We conducted the following studies:

A) We studied the difference between killing 10% of cells in young and old tissue as follows:

Killing cells at random across the tissue using mechanism 1a) (purse-string closure after cell apoptosis).

Killing clusters of cells across the tissue using mechanism 1b) (purse-string closure after a cluster of cell apoptosis).

B) We analyzed cell apoptosis in isolated small and medium drusen and compared it with experimental measurements. Cell apoptosis was simulated in three different ways:

Purse-string closure after cell apoptosis, mechanism 1a).

Cell fusion, mechanism 2a).

Cell expansion, mechanism 3a).

C) We killed clusters of cells according to mechanism 1b) and compared the results with tissues exhibiting AMD.

D) We killed cells according to mechanism 2c) and compared the results with tissues exhibiting AMD.

For data analysis, we used different statistical tools depending on the cases:

Area distributions

Number of neighbor cell distributions

Voronoi index

Texture tensor

Spatial point analysis using cumulative nearest neighbor distance distribution function

Second-order correlation of the number of cells

We use Fogbank for individual cell segmentation of RPE tissues